Lecture 12 – Garbage Collection

COSE212: Programming Languages

Jihyeok Park



2026 Fall

Recall

- Mutation makes it possible to **change the state** of a program by **updating the contents** of a data structure or a variable.
  - BFAE – FAE with **mutable boxes**
  - MFAE – FAE with **mutable variables**
  - Evaluation with **memories**, finite maps from addresses to values:

$$\begin{array}{ll} \text{Memories} & M \in \mathbb{A} \xrightarrow{\text{fin}} \mathbb{V} \\ \text{Addresses} & a \in \mathbb{A} \end{array}$$

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- In this lecture, we will learn **memory management** techniques to **deallocate** unreachable memory cells:
  - **Stack and Heap**
  - **Manual Memory Management**
  - **Garbage Collection (GC)**

Contents

1. Stack and Heap
    - Tail-Call Optimization (TCO)

2. Manual Memory Management

3. Garbage Collection
    - Reference Counting
    - Mark-and-Sweep GC
    - Copying GC (Two-Space GC)
    - Other Techniques

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Stack and Heap

In the previous lecture, we saw that the memory of the following MFAE expression has **unreachable** memory cells:

```
/* MFAE */
var y = 1;
var f = x => {
  x = x + y;
  x * x
};
f(5);          /* 36 */
y = 3;
f(5)           /* 64 */        *
```

$$\sigma = [\; \mathtt{y} \mapsto a_0,\; \mathtt{f} \mapsto a_1 \;]$$

$\mathbb{A}$:	a_0	a_1	a_2	a_3	...
M =	3	v	6	8	...

where $v = \langle \lambda \mathtt{x}.(\mathtt{x = x + y;\ x * x}), [\mathtt{y} \mapsto a_0] \rangle$

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Let's delete **unreachable** memory cells when the program **exits** functions!

Stack and Heap

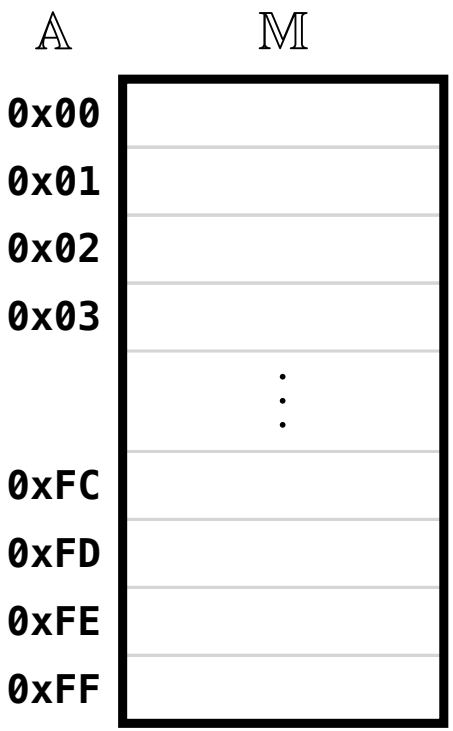
We can **divide** the memory into two parts:

- **Stack** – for **local variables** and **function parameters**
- **Heap** – for **dynamically allocated** memory cells

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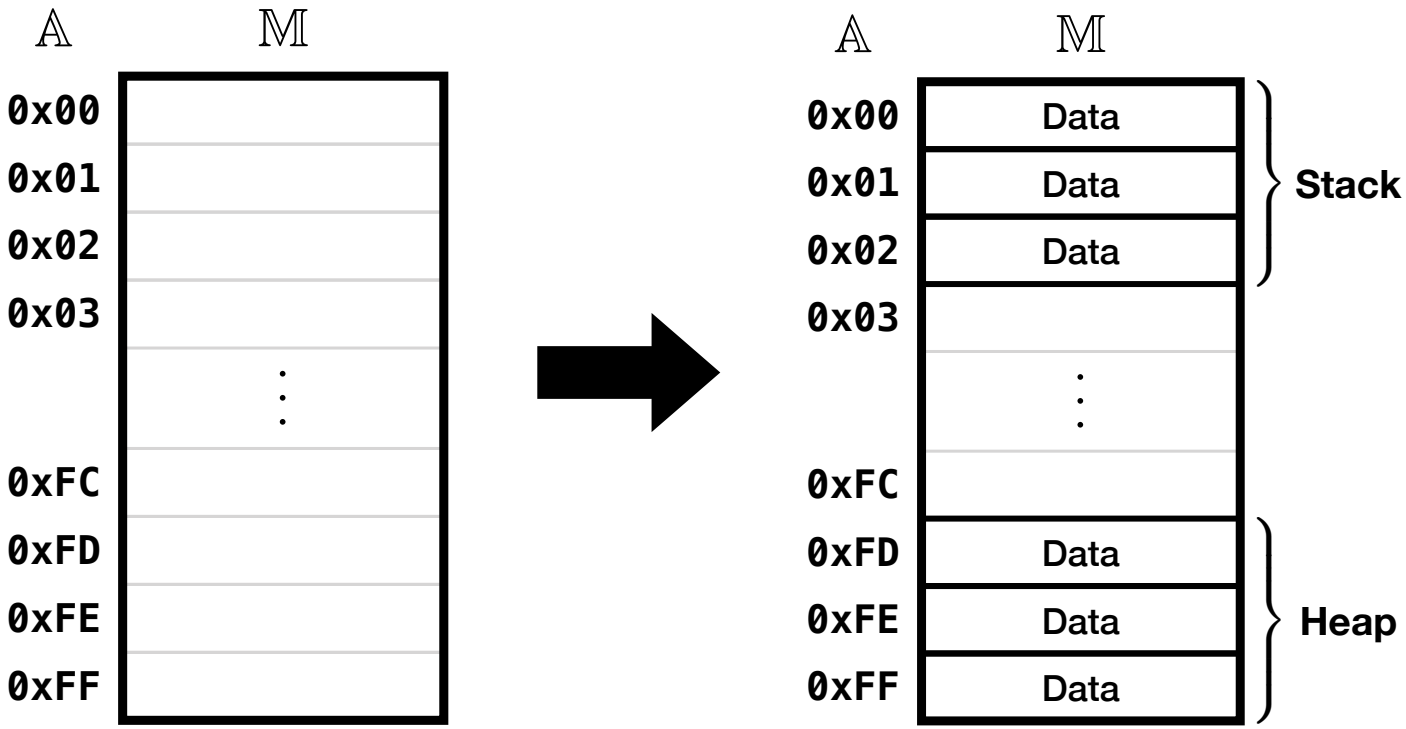
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Stack and Heap

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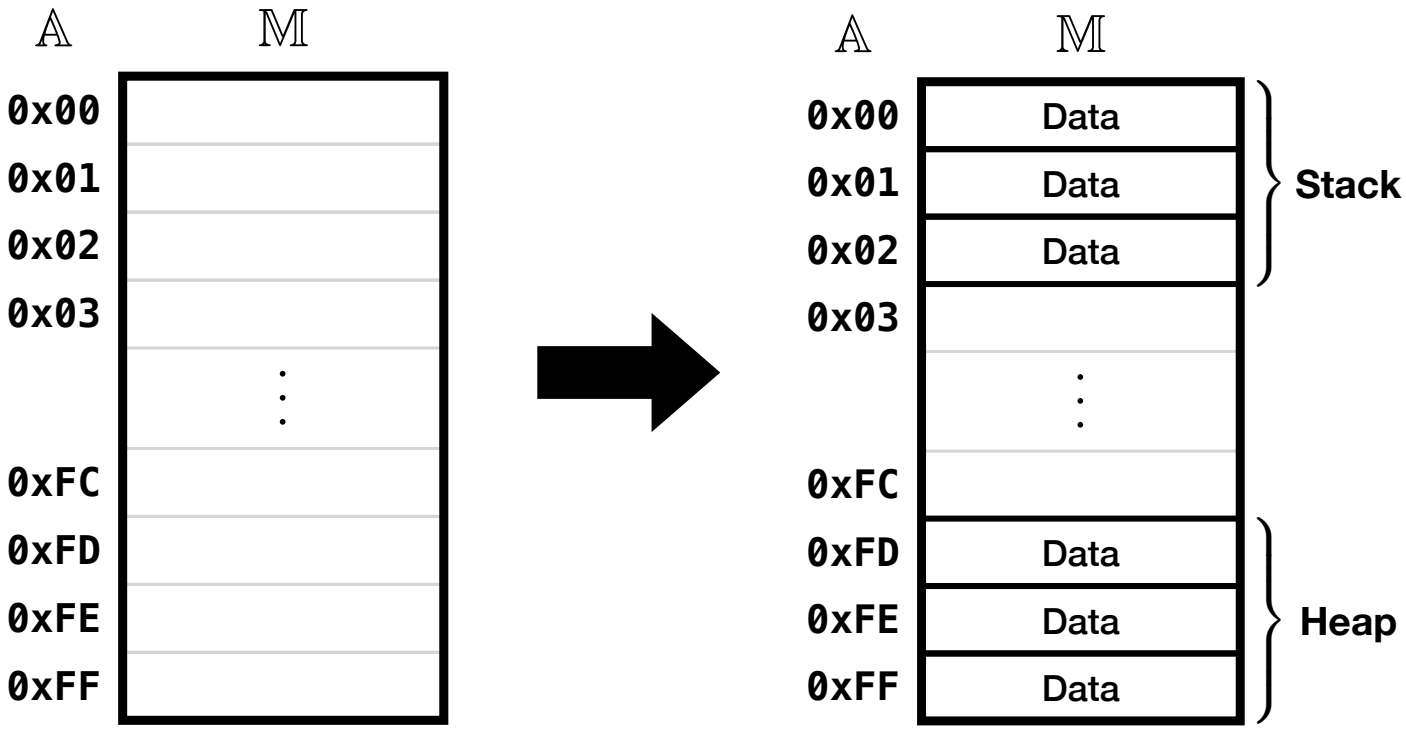
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Stack and Heap

We can **divide** the memory into two parts:

- **Stack** – for **local variables** and **function parameters**
- **Heap** – for **dynamically allocated** memory cells



Create a **new stack frame** when the program **enters** a function, and **delete** the stack frame when it **exits** the function.

Stack and Heap

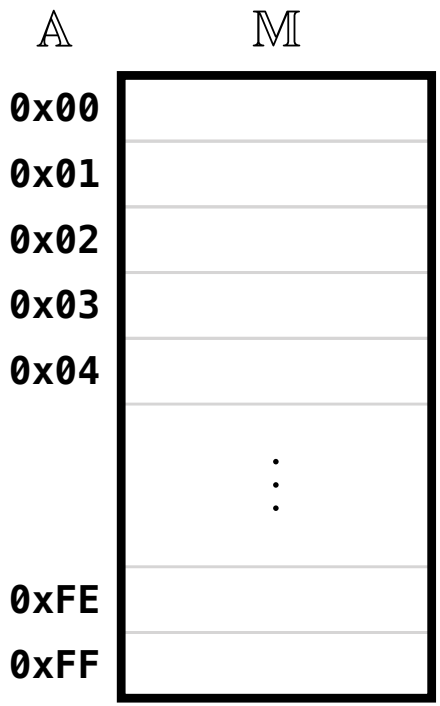
For example, consider the following Scala program:

```
case class Box(var k: Int)

def f(x: Int): Int =
  var y = Box(1)
  var z = g(2)
  x + y.k + z

def g(b: Int): Int =
  var c = Box(b)
  c.k + 3

var a = 1
var b = f(42)
a + b
```



Stack and Heap

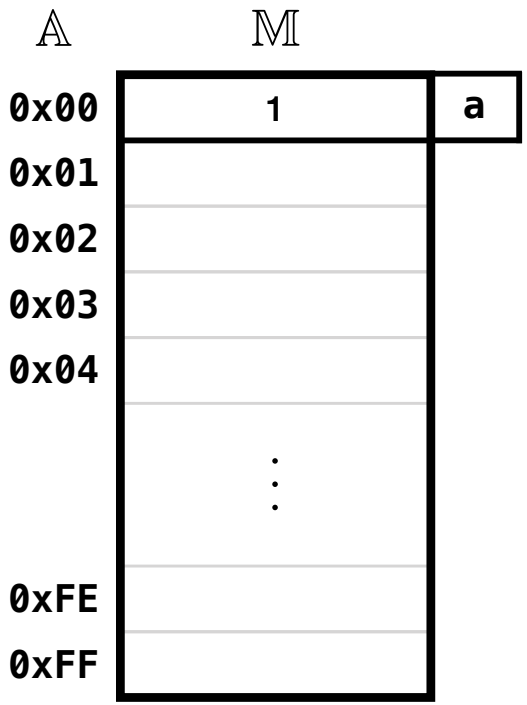
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Stack and Heap

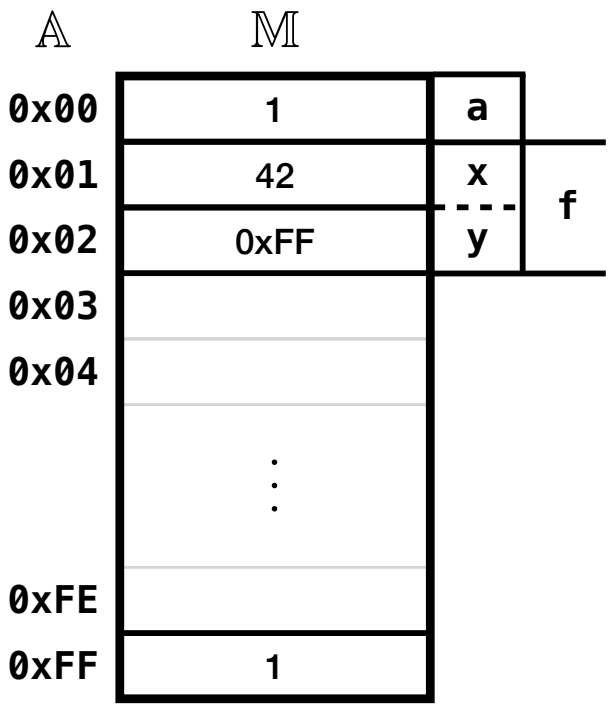
For example, consider the following Scala program:

```
case class Box(var k: Int)

def f(x: Int): Int = /* x -> 0x01 */
  var y = Box(1)     /* y -> 0x02 */  *
  var z = g(2)
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def g(b: Int): Int =
  var c = Box(b)
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A **new stack frame** is created when the program enters the function `f`.

Stack and Heap

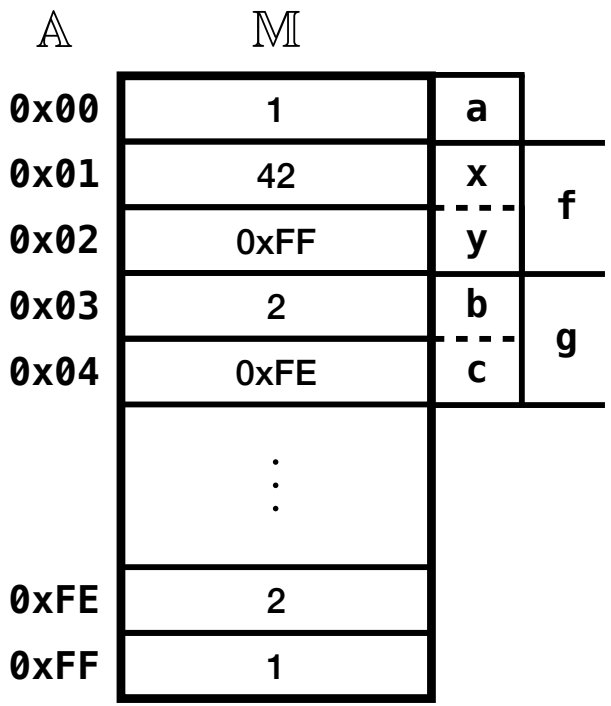
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def g(b: Int): Int = /* b -> 0x03 */
  var c = Box(b)     /* c -> 0x04 */
  c.k + 3            /* 5 */          *

var a = 1            /* a -> 0x00 */
var b = f(42)
a + b
```



A **new stack frame** is created when the program enters the function g.

Stack and Heap

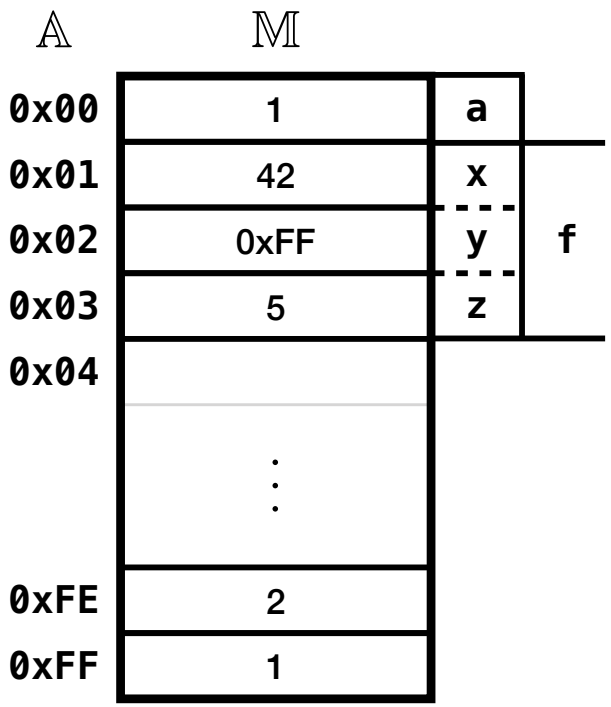
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  x + y.k + z        /* 48 */

def g(b: Int): Int =
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var a = 1            /* a -> 0x00 */
var b = f(42)
a + b
```



After exiting the function g, its stack frame is **deleted**. The memory cells allocated for b and c in the stack frame are **deallocated**.

Stack and Heap

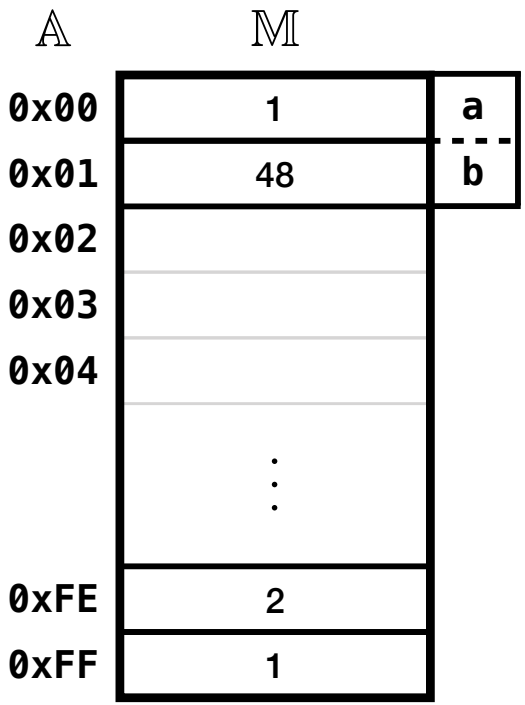
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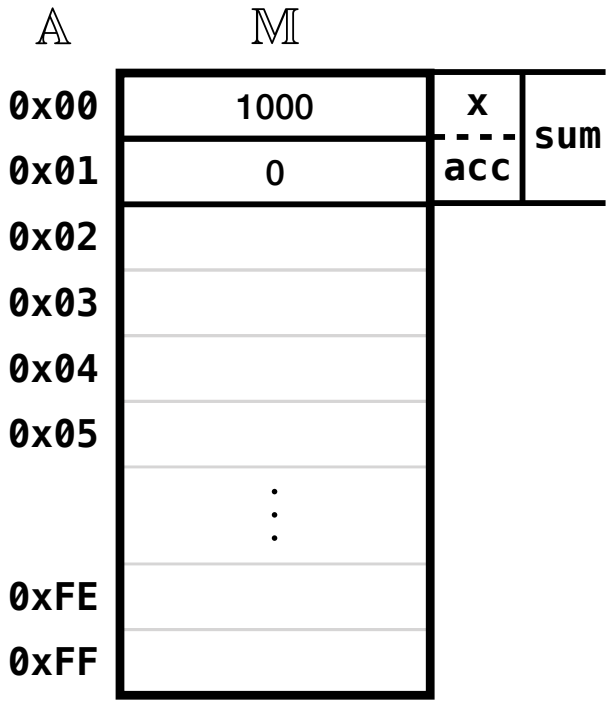
After exiting the function `f`, its stack frame is **deleted**. The memory cells allocated for `x`, `y`, and `z` in the stack frame are **deallocated**.

Tail-Call Optimization (TCO)

Here is another example with a recursive function:

```
def sum(x: Int, acc: Int): Int =
  if (x < 1) acc
  else sum(x - 1, x + acc)

sum(1000, 0)
```

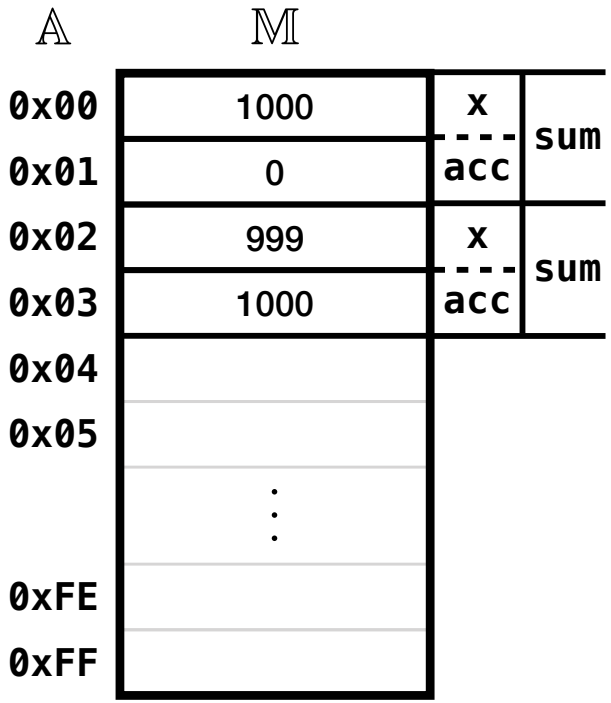


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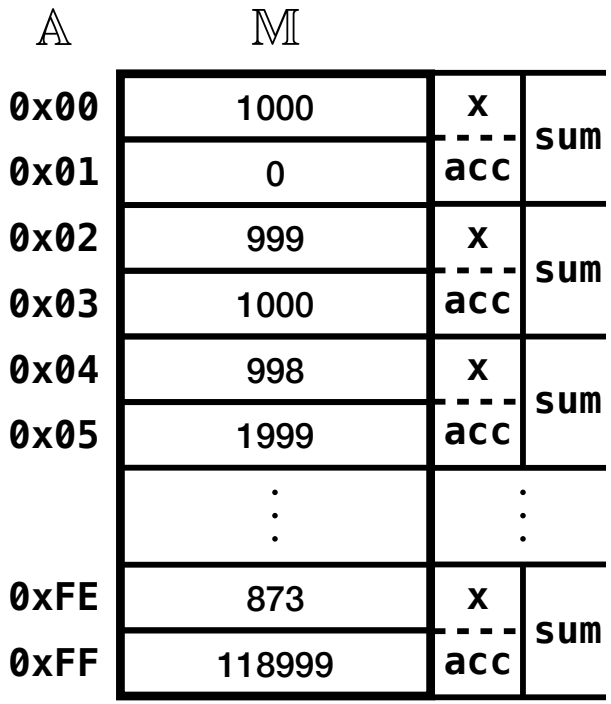


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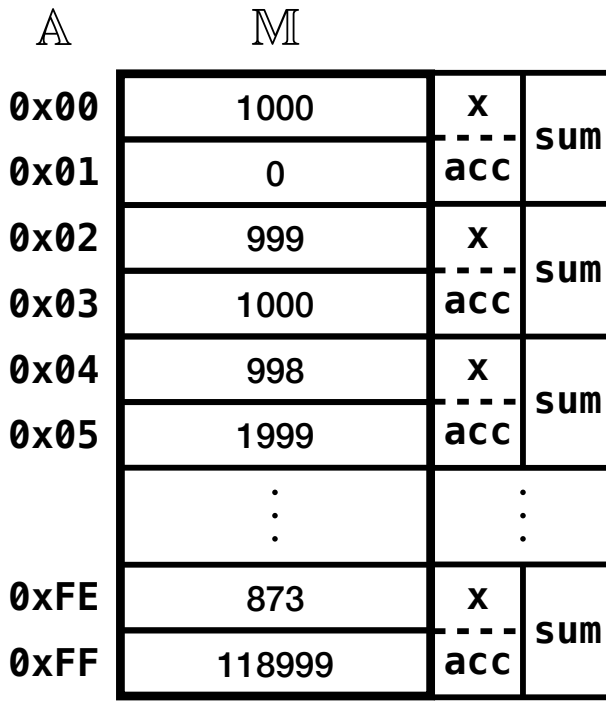
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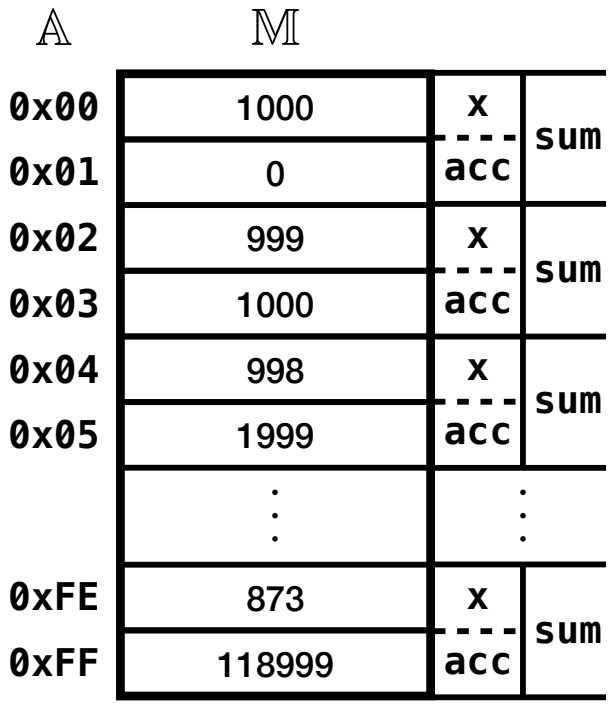
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It fails with a **stack overflow** error.

However, is it really necessary to keep **all the stack frames**? **No!**

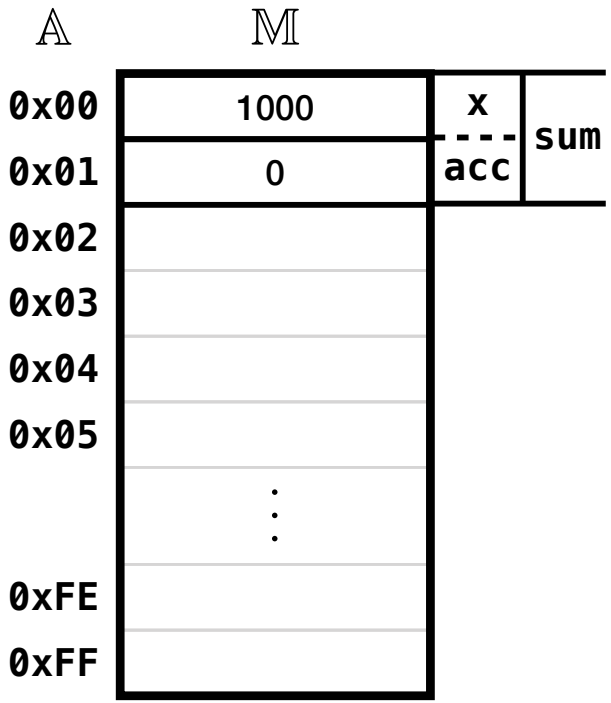
Scala supports **tail-call optimization** (TCO).

Tail-Call Optimization (TCO)

Here is another example with a recursive function:

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def sum(x: Int, acc: Int): Int =
  if (x < 1) acc
  else sum(x - 1, x + acc) // tail-call

sum(1000, 0)
```



Why? The function call is in the **tail-call position** (i.e., the final action in the function). It means that it directly returns the result without any further computation.

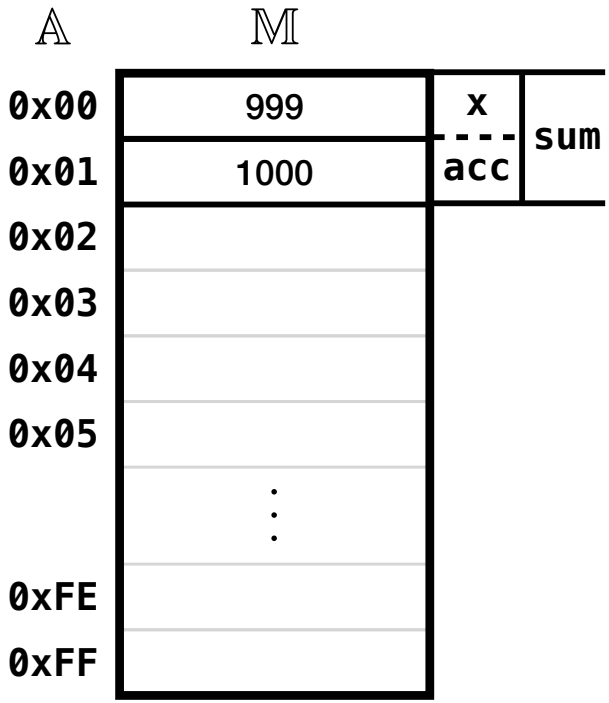
Thus, we can safely **discard** the current stack frame **before** calling the function, and it is called **tail-call optimization** (TCO).

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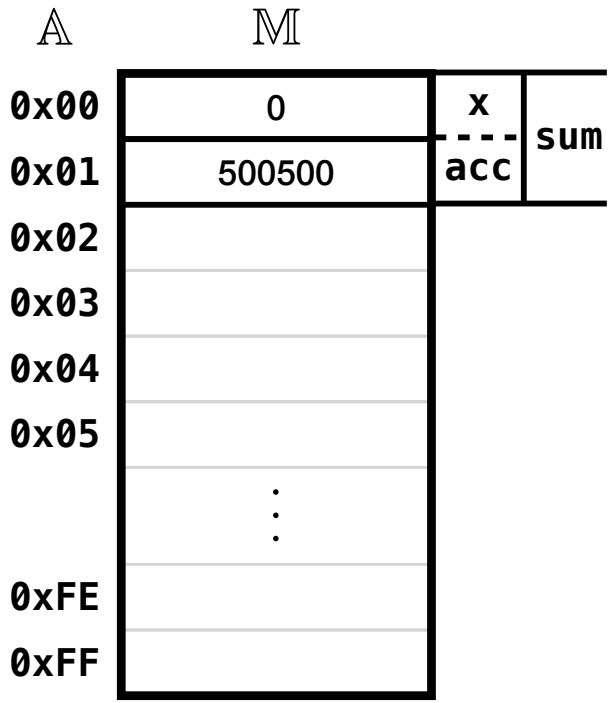
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sum(1000, 0) // 500500
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Thus, we can safely **discard** the current stack frame **before** calling the function, and it is called **tail-call optimization** (TCO).

Tail-Call Optimization (TCO)

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  if (x < 2) 1
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After `factorial(x - 1)`, it still needs to multiply the result by `x`.

Then, how can we put it in the **tail-call position**?

Tail-Call Optimization (TCO)

PLRG

One common pattern for TCO is to use an **accumulator**:

```
def factorial(x: Int, acc: Int): Int =
  if (x < 2) acc
  else factorial(x - 1, x * acc)

factorial(5, 1) // 120
```

Tail-Call Optimization (TCO)

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factorial(5, 1) // 120
```

However, it is not a user-friendly interface because we need to pass the initial value of the accumulator (e.g., 1) every time.

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def factorial(x: Int, acc: Int): Int =
  if (x < 2) acc
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factorial(5, 1) // 120
```

However, it is not a user-friendly interface because we need to pass the initial value of the accumulator (e.g., 1) every time. We can define a nested function to hide the additional parameter:

```
def factorial(x: Int): Int =
  def aux(x: Int, acc: Int): Int =
    if (x < 2) acc
    else aux(x - 1, x * acc)
  aux(x, 1)

factorial(5) // 120
```

Tail-Call Optimization (TCO)

Many programming languages support **tail-call optimization** (TCO) to avoid stack overflow errors.

Tail-Call Optimization (TCO)

PLRG

Many programming languages support **tail-call optimization** (TCO) to avoid stack overflow errors.

In addition, Scala supports the `@tailrec` annotation to check at compile time whether a recursive call is in the tail-call position:

```
import scala.annotation.tailrec

// Passes the tail-call position check
@tailrec
def factorial(x: Int, acc: Int): Int =
  if (x < 2) acc
  else factorial(x - 1, x * acc)

// Compile-time error
@tailrec
def factorial(x: Int): Int =
  if (x < 2) 1
  else x * factorial(x - 1)
```

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Manual Memory Management

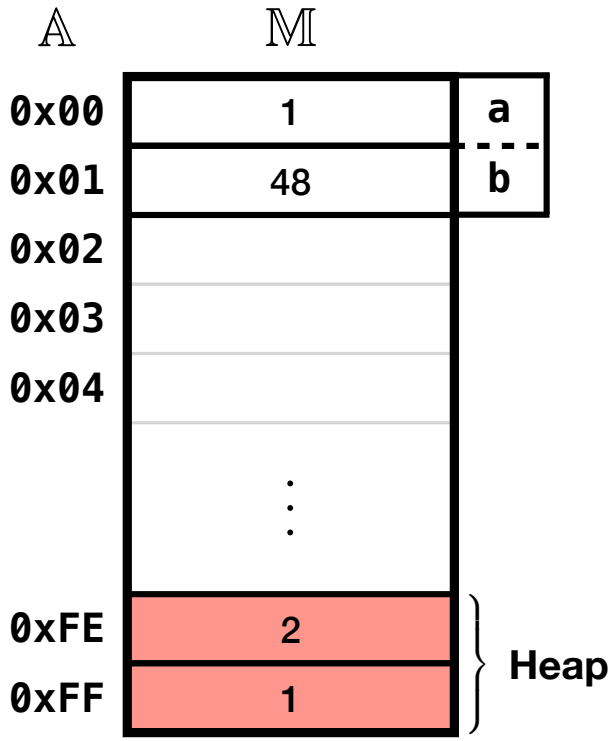
Let's see the previous example again:

```
case class Box(var k: Int)

def f(x: Int): Int =
  var y = Box(1)
  var z = g(2)
  x + y.k + z

def g(b: Int): Int =
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var a = 1                /* a -> 0x00 */
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a + b                    /* 49 */          *
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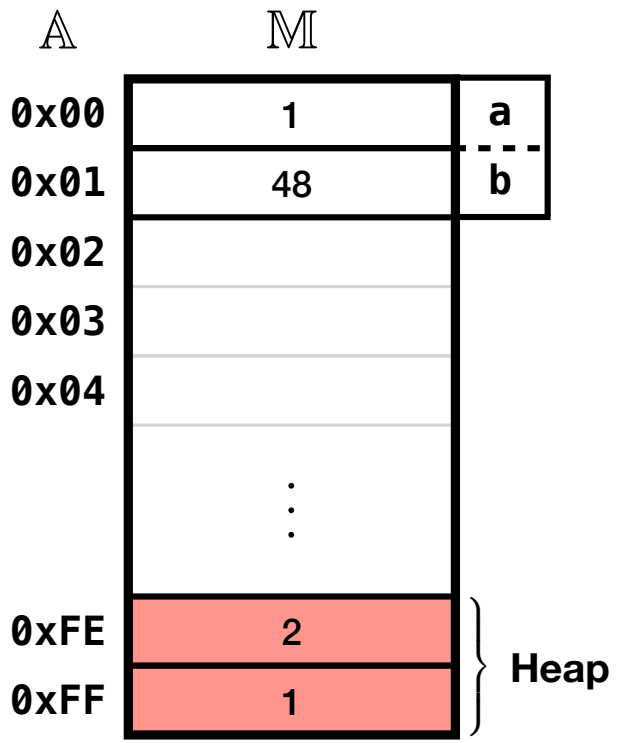


Unfortunately, we still cannot deallocate memory cells (e.g., `0xFE` and `0xFF`) dynamically allocated in the **heap** rather than the **stack**.

Manual Memory Management

One way to resolve this is **manual memory management**: e.g., C++ provides keywords for allocation (`new`) and deallocation (`delete`):

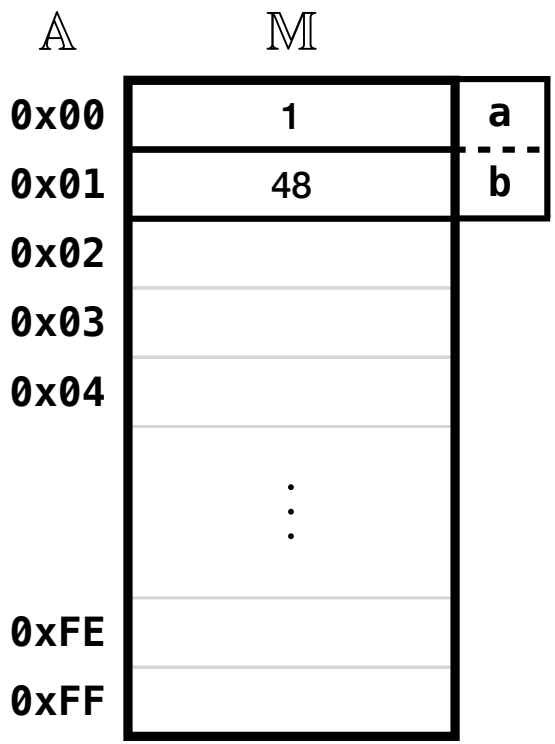
```
struct Box { int k; Box(int k): k(k) {} };
int f(int x) {
  Box* y = new Box(1);     // alloc   0xFF
  int z = g(2);
  int k = y->k;
  return x + k + z;
}
int g(int b) {
  Box* c = new Box(b);     // alloc   0xFE
  int k = c->k;
  return k + 3;
}
int a = 1;      /* a -> 0x00 */
int b = f(42); /* b -> 0x01 */
a + b;          /* 49 */          *
```



Manual Memory Management

One way to resolve this is **manual memory management**: e.g., C++ provides keywords for allocation (`new`) and deallocation (`delete`):

```
struct Box { int k; Box(int k): k(k) {} };
int f(int x) {
  Box* y = new Box(1);     // alloc   0xFF
  int z = g(2);
  int k = y->k; delete y; // dealloc 0xFF
  return x + k + z;
}
int g(int b) {
  Box* c = new Box(b);     // alloc   0xFE
  int k = c->k; delete c; // dealloc 0xFE
  return k + 3;
}
int a = 1;      /* a -> 0x00 */
int b = f(42); /* b -> 0x01 */
a + b;          /* 49 */          *
```



Manual Memory Management

PLRG

Pros:

- **Efficient** – Users can **explicitly** deallocate memory cells allocated in heap whenever they want.

Cons:

- **Error-prone** – Users have all the **responsibility** to deallocate memory cells allocated in heap:
  - **Memory Leak** occurs if users forget to deallocate memory cells.

```
b = new Box(42);  ...
```

  - **Use After Free (UAF)** occurs if users try to access memory cells already deallocated.

```
b = new Box(42);  ...  delete b;  ...  b->k;
```

  - **Double Free** occurs if users deallocate memory cells more than once.

```
b = new Box(42);  ...  delete b;  ...  delete b;
```

Contents

1. Stack and Heap
    Tail-Call Optimization (TCO)

2. Manual Memory Management

3. Garbage Collection
    Reference Counting
    Mark-and-Sweep GC
    Copying GC (Two-Space GC)
    Other Techniques

Garbage Collection

Is there any way to **automatically** deallocate memory cells in heap?

Garbage Collection

Is there any way to **automatically** deallocate memory cells in heap? **Yes!**

Garbage collection (GC) is a representative technique for **automatic memory management** that automatically **detects** and **deallocates** memory cells that are no longer reachable in heap.

Let's learn several GC algorithms:

- **Reference counting**
- **Mark-and-sweep GC**
- **Copying GC** (Two-space GC)
- Others

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PLRG

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Let's learn several GC algorithms:

- **Reference counting**
- **Mark-and-sweep GC**
- **Copying GC** (Two-space GC)
- Others

Before explaining them, let's represent memory cells in heap in a **graphical way** without actual addresses.

Garbage Collection

From now on, we will use the **graphical representation** of memory cells:

```
case class Box(var x: Box)

var a = Box(Box(Box(null)))  *

var b = Box(Box(null))

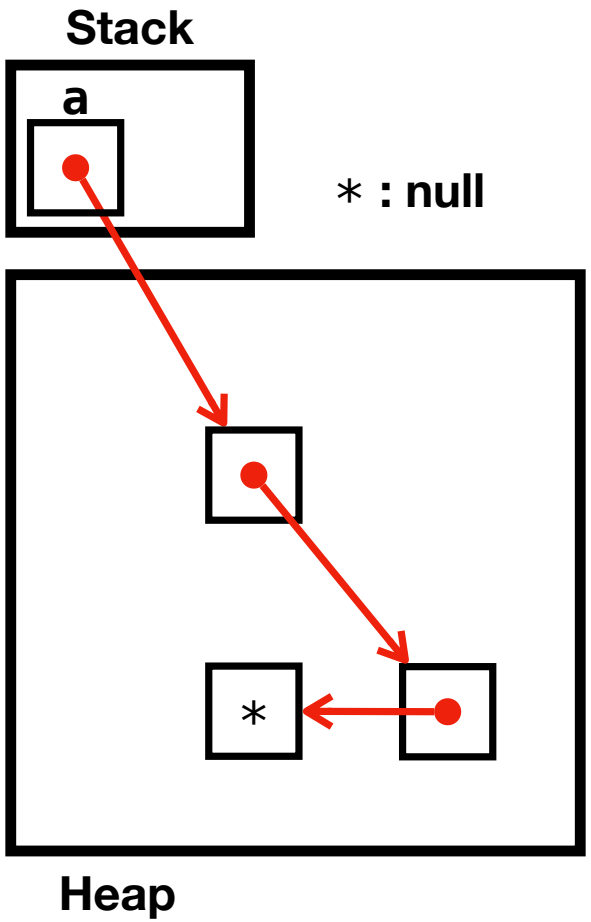
a.x.x.x = a.x

b.x = a

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a = a.x

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```



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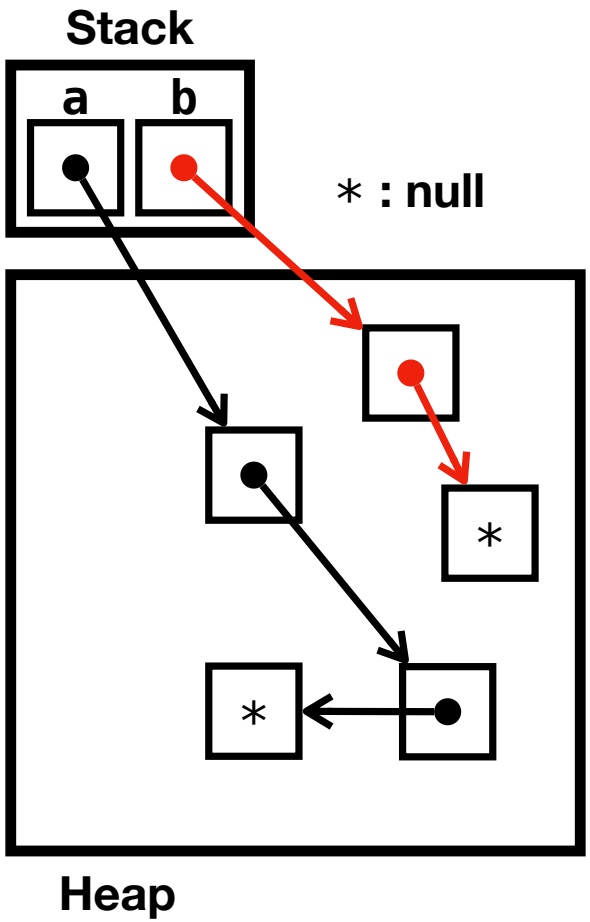
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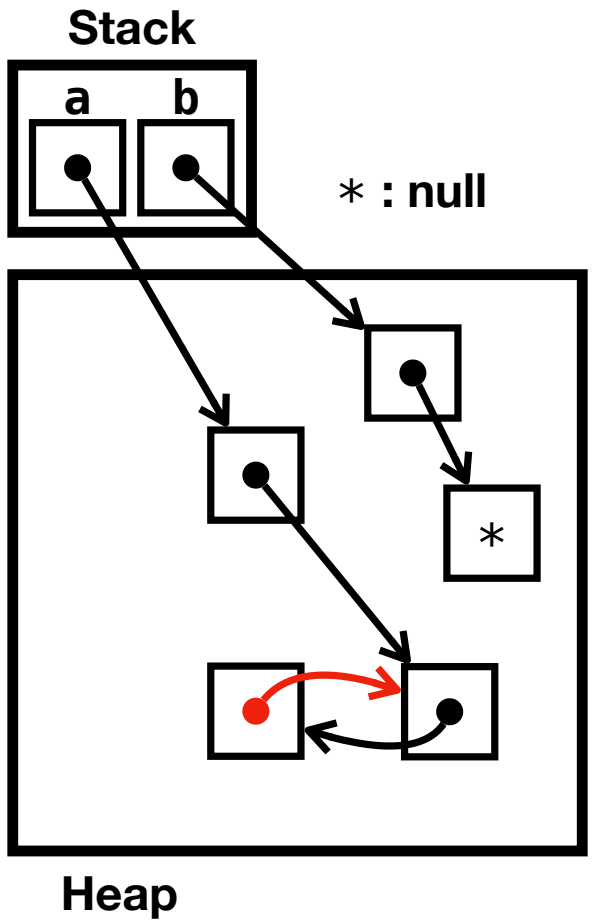
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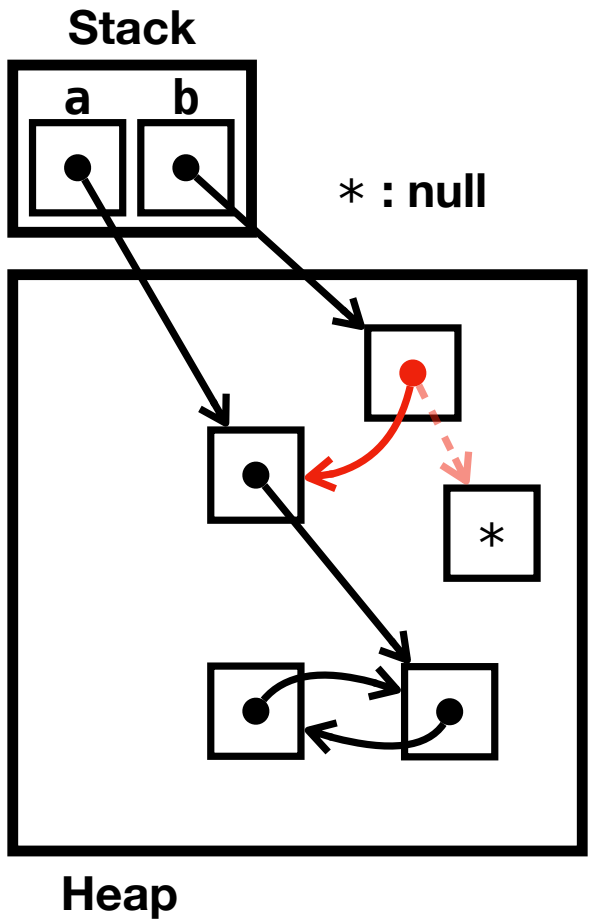
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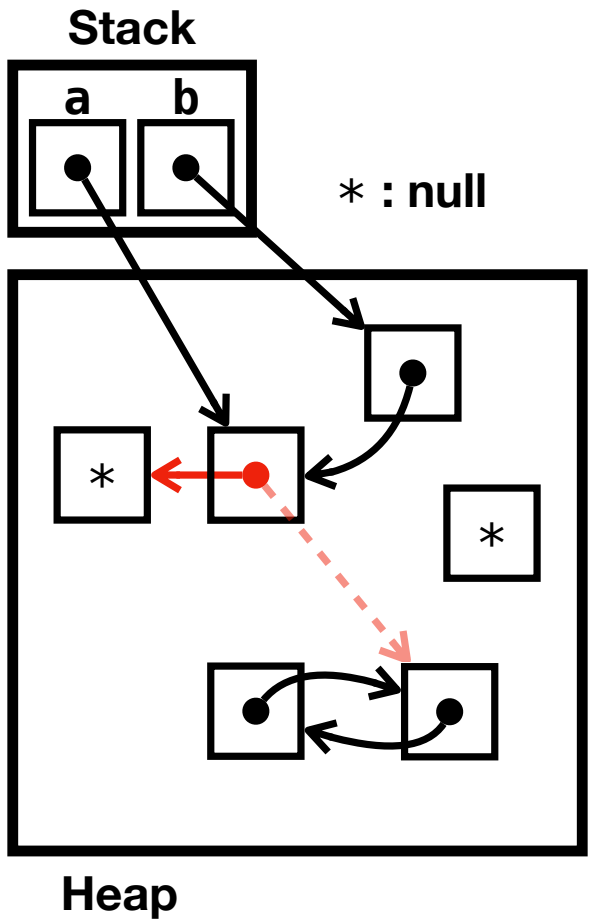
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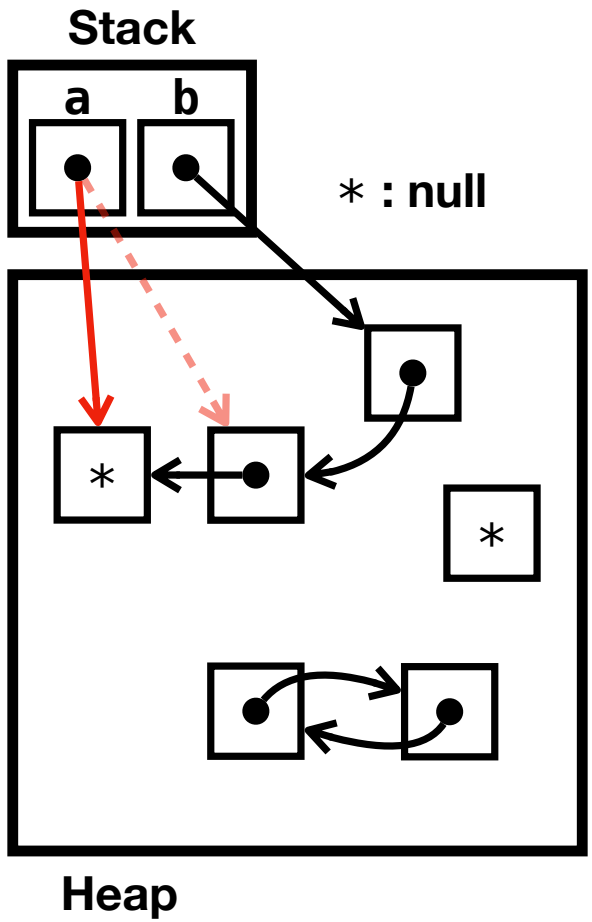
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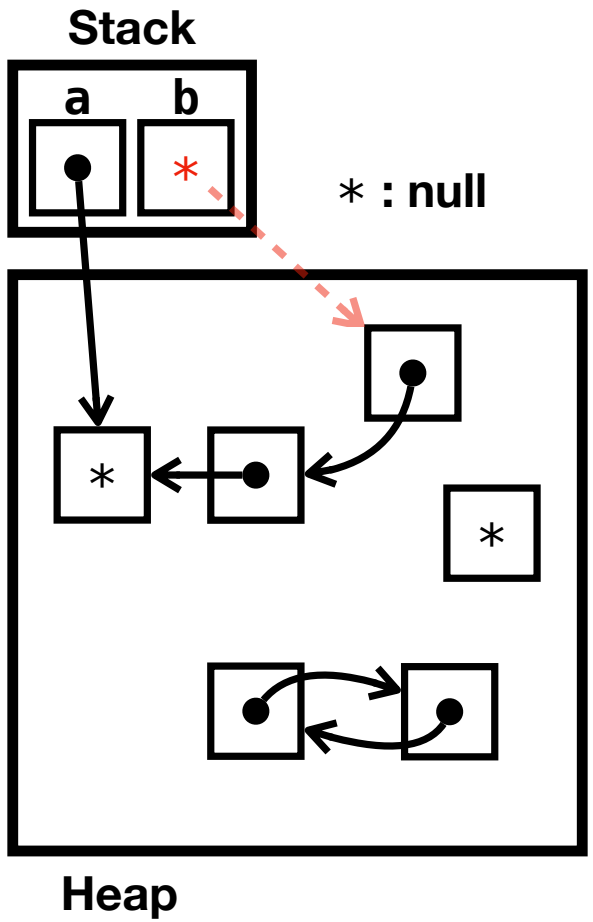
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a = a.x

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```



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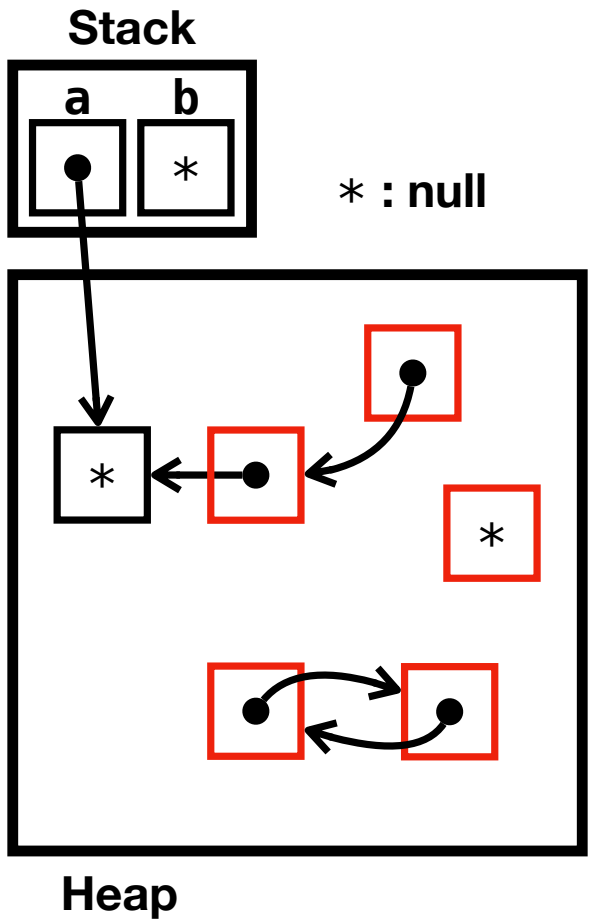
a.x.x.x = a.x

b.x = a

a.x = Box(null)

a = a.x

b = null                    *
```



We need to deallocate five unreachable memory cells in heap.

Reference Counting

Reference counting is a simple GC algorithm that keeps track of the **number of references** to each memory cell in heap.

1. **Initialize** the reference count of each cell to 0.
2. When a reference to a cell is created, **increment** its reference count.
3. When a reference to a cell is deleted, **decrement** its reference count.
4. When the reference count of a cell reaches 0, **deallocate** the cell.

Many programming languages use reference counting to implement GC:

- Python, Swift, Perl, Objective-C, etc.

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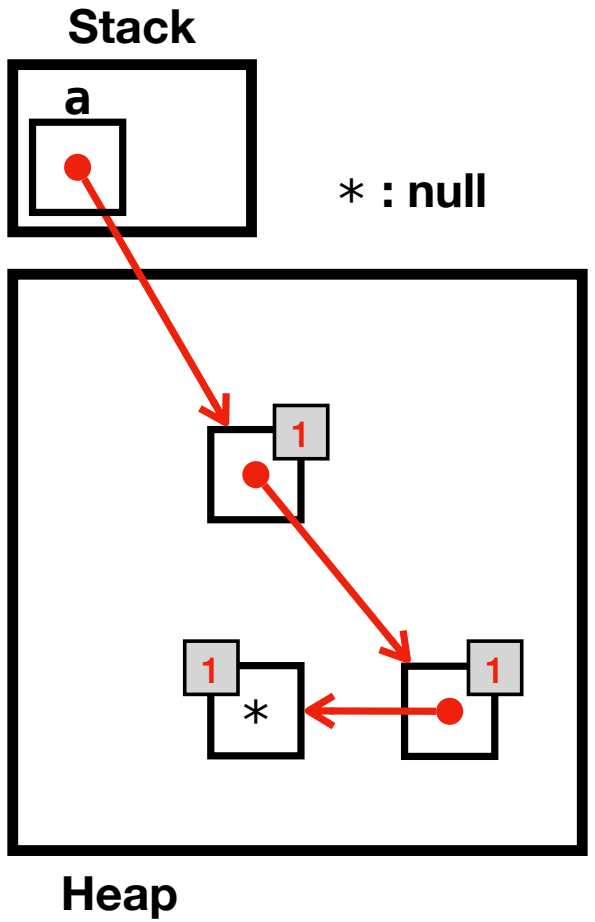
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b.x = a

a.x = Box(null)

a = a.x

b = null
```



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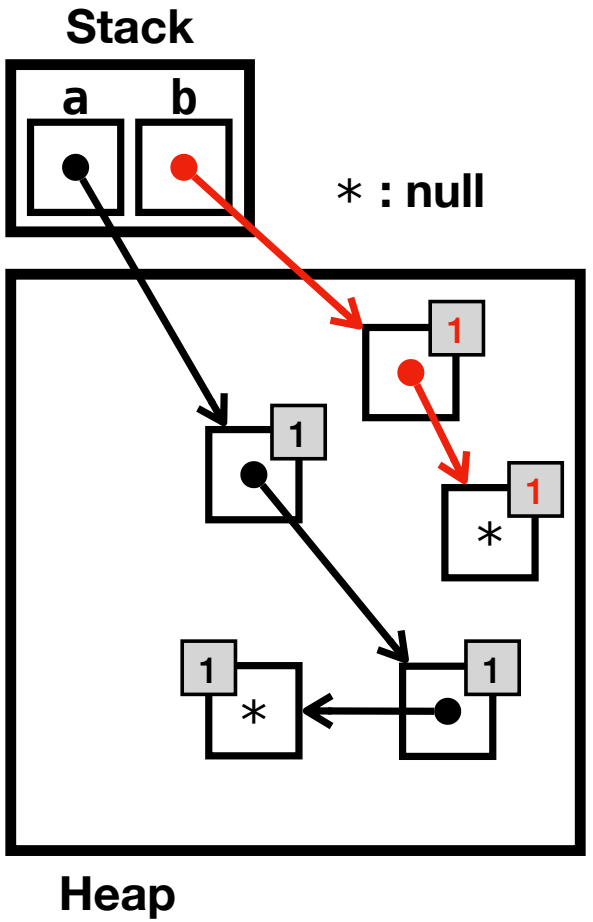
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b.x = a

a.x = Box(null)

a = a.x

b = null
```



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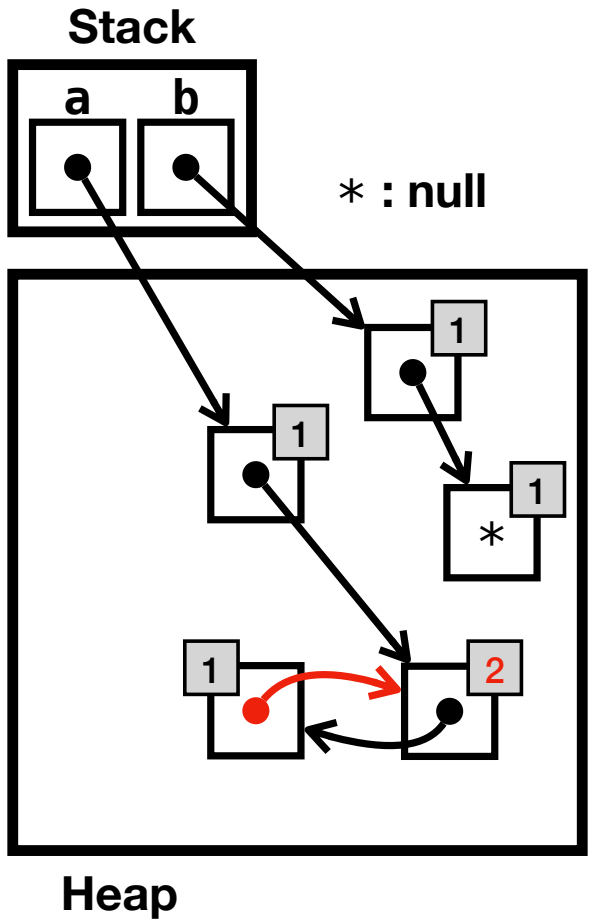
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a = a.x

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```



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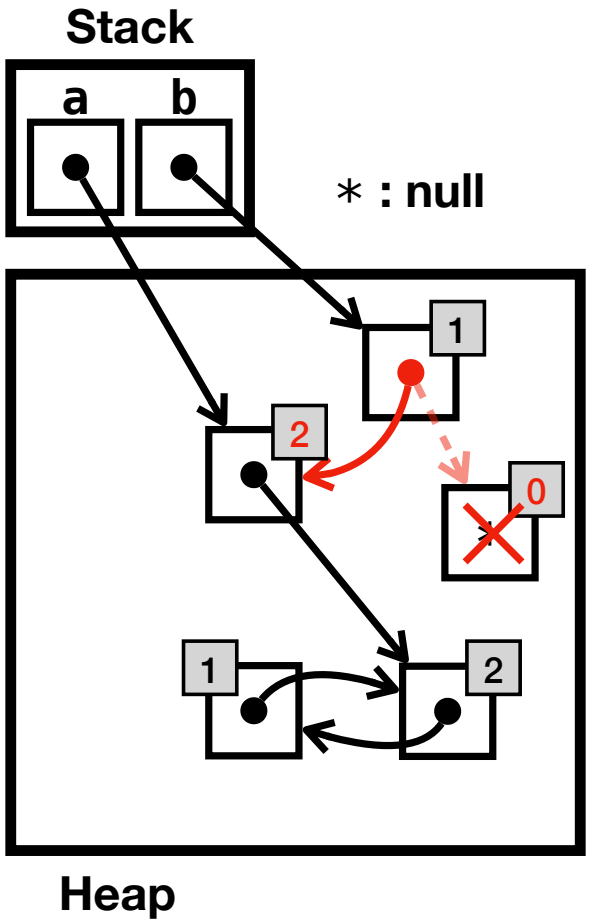
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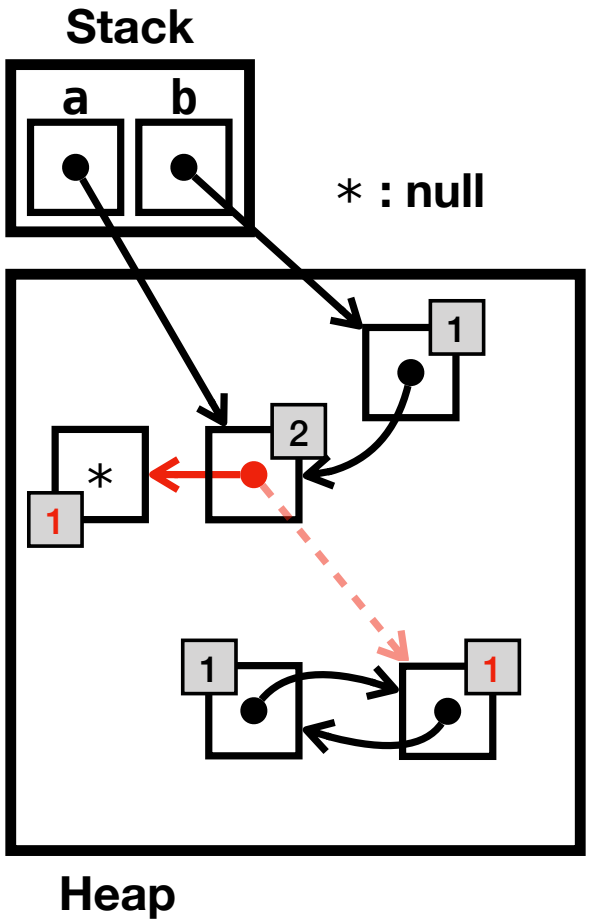
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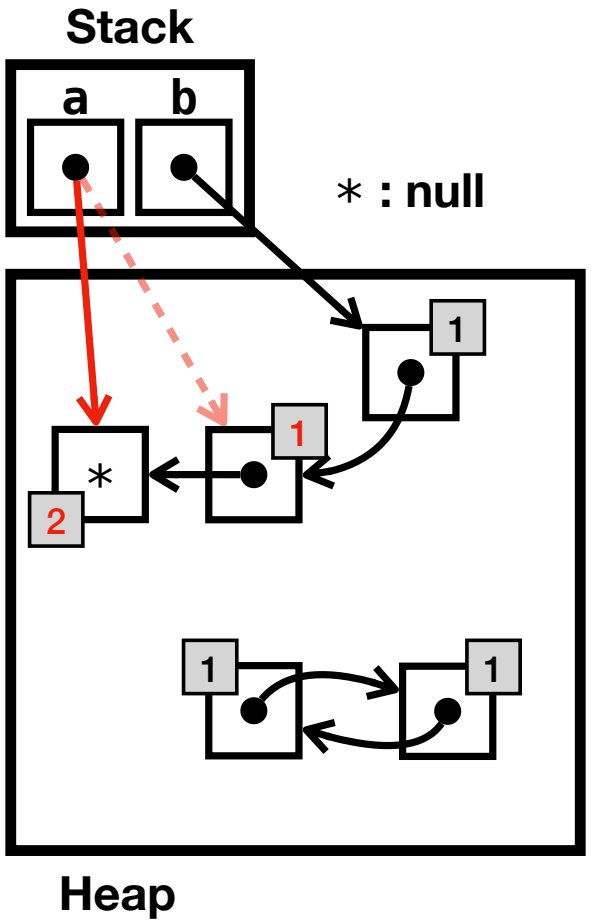
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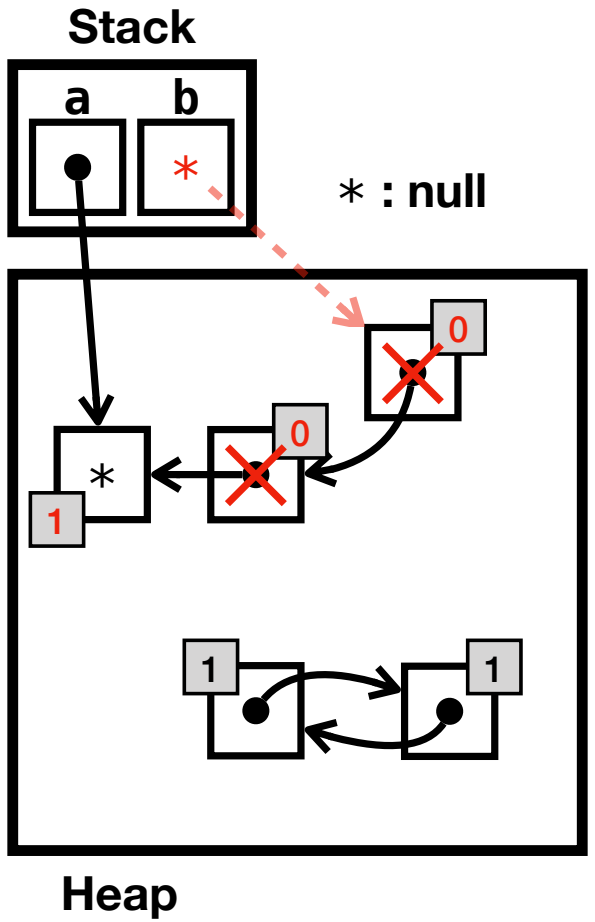
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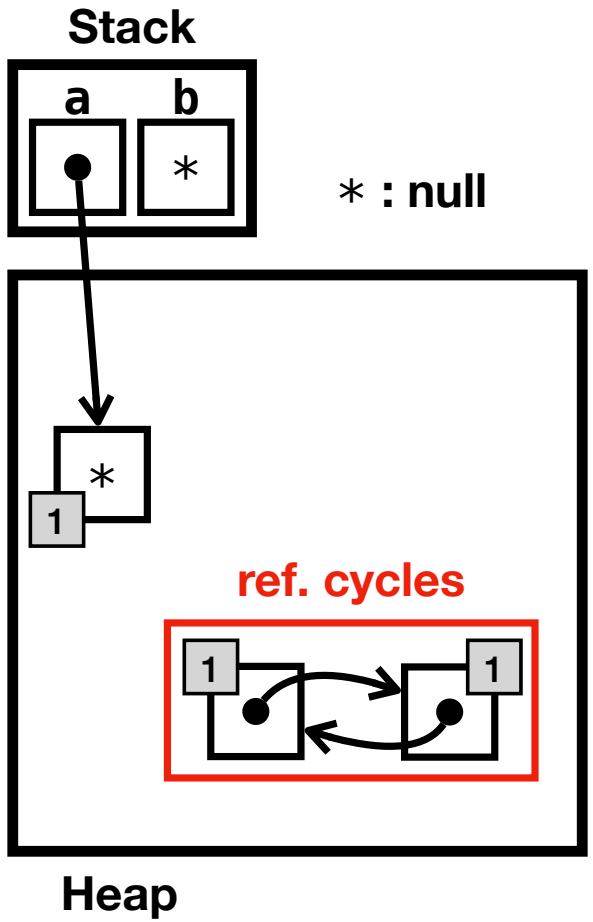
a.x.x.x = a.x

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b = null                    *
```



Unfortunately, we cannot deallocate unreachable **reference cycles**.

Reference Counting

Reference counting is a simple GC algorithm that keeps track of the **number of references** to each memory cell in heap.

Pros:

- **Easy to implement** – Simply increment and decrement the reference count when a reference is created and deleted.
- **No pauses** – Deallocation happens incrementally and immediately, without **stop-the-world** pauses.

Cons:

- **Reference cycles** – It cannot deallocate unreachable reference cycles.
- **Reference count cost** – It requires extra space and time to maintain reference counts.
- **Free List** and **Fragmentation** – It requires a **free list** to keep track of available free memory cells in heap, and it also suffers from **fragmentation** making it difficult to allocate large objects.

Mark-and-Sweep GC

Mark-and-Sweep GC is one of the **tracing GC** algorithms that **traverses the heap** to find unreachable objects when it is **triggered** under some conditions.

1. **Mark** all memory cells as **unreachable** (white).
2. **Mark** all memory cells referenced by roots as **unscanned** (gray).
3. Repeat until there are no unscanned (gray) memory cells:
   1. **Pick** an unscanned (gray) memory cell.
   2. **Mark** memory cells referenced by the picked one as **unscanned** (gray).
   3. **Mark** the picked memory cell as **scanned** (black).
4. **Deallocate** (sweep) all memory cells that are still marked as **unreachable** (white).

Mark-and-Sweep GC

Assume that the GC is triggered at the following program point, and let's perform **mark-and-sweep** GC.

```
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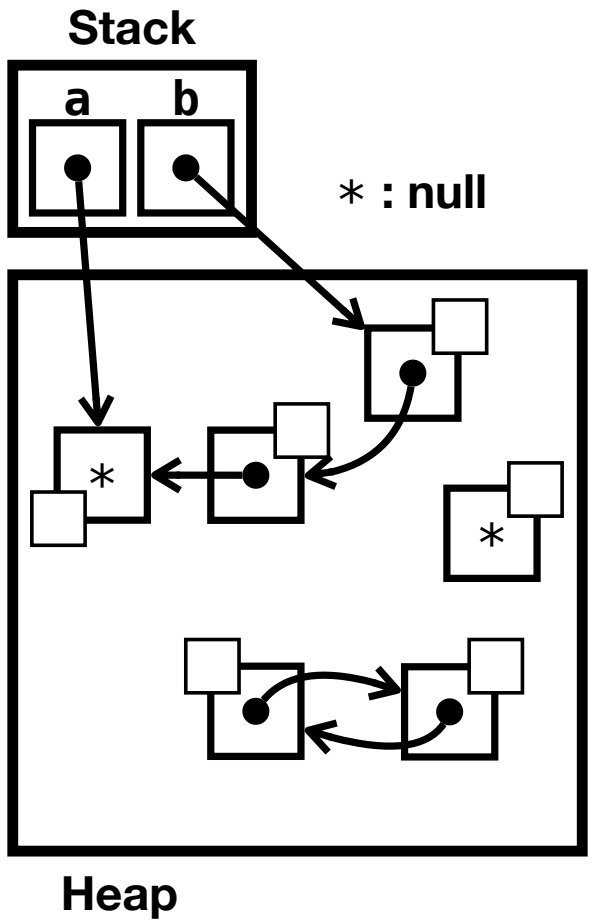
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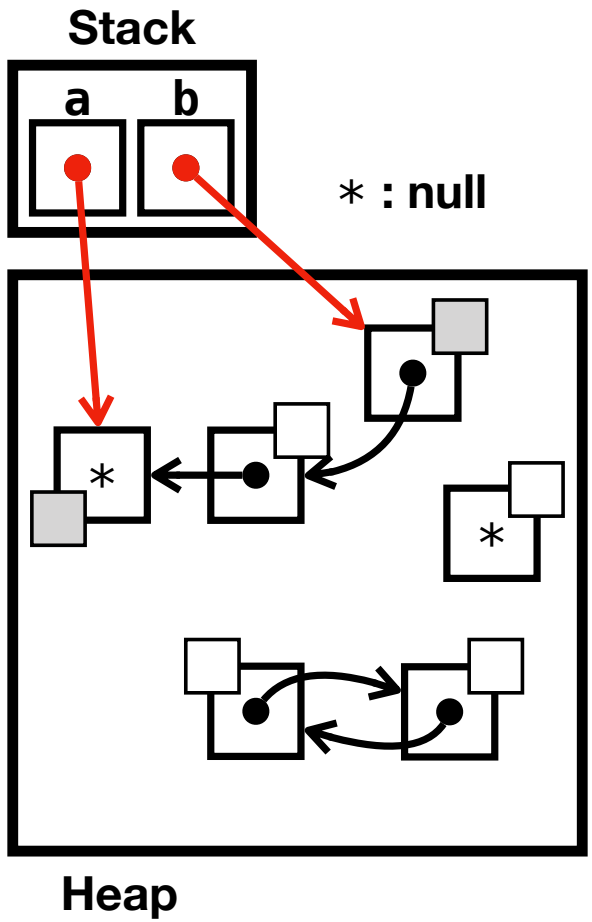
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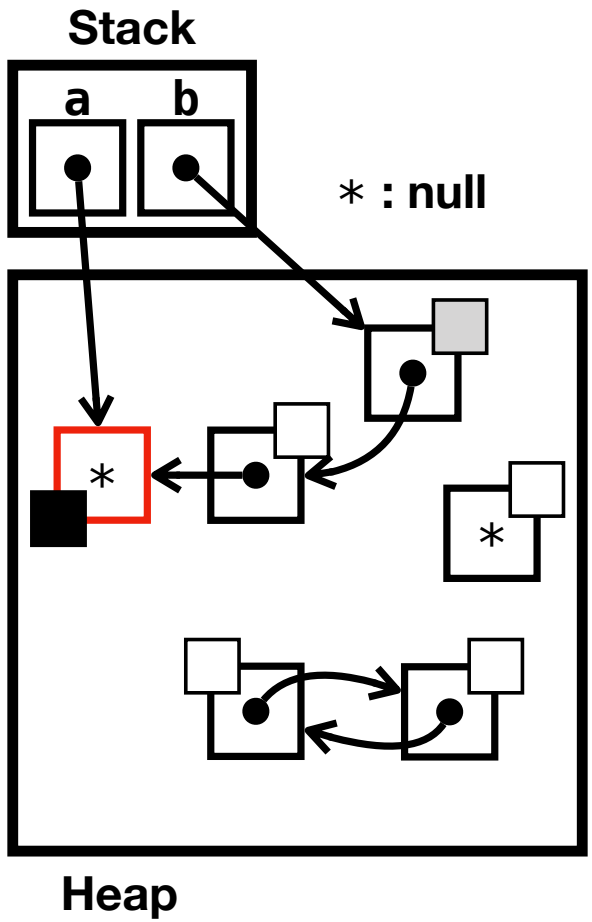
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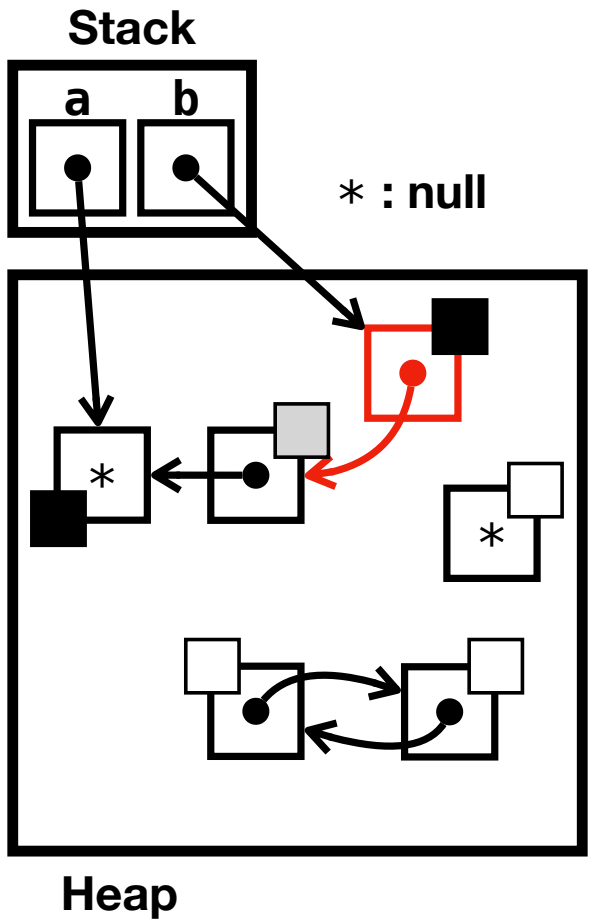
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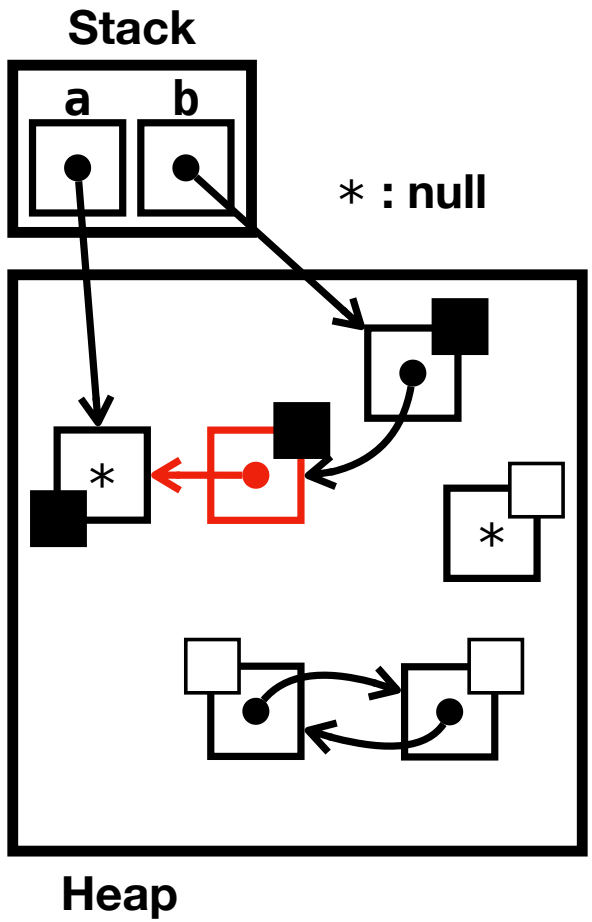
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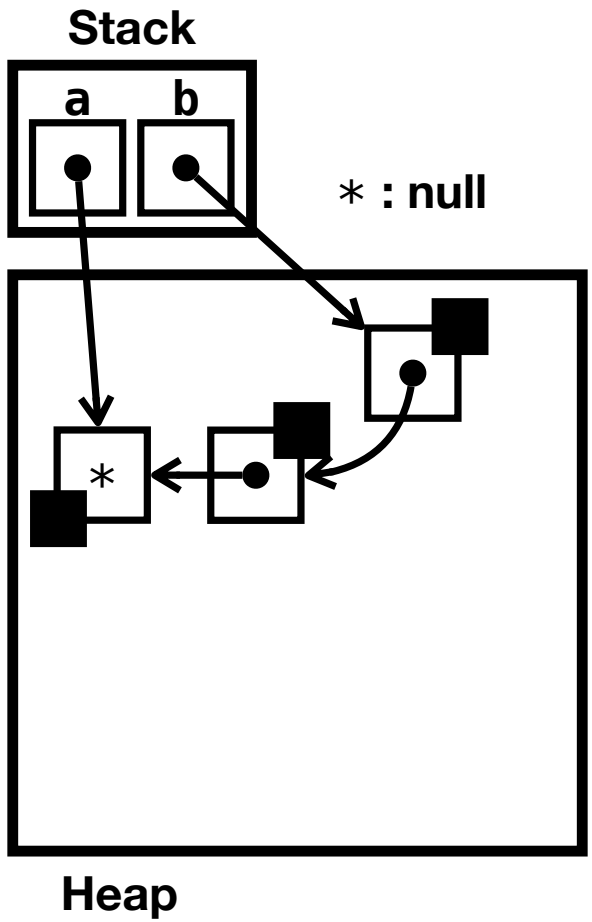
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Mark-and-Sweep GC

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Pros:

- **Handle Reference Cycles** – It can deallocate unreachable reference cycles.

Cons:

- **Stop-the-world** – It **stops** the program execution during GC.
- **Free List** and **Fragmentation** – It requires a **free list** to keep track of available free memory cells in heap, and it also suffers from **fragmentation** making it difficult to allocate large objects.

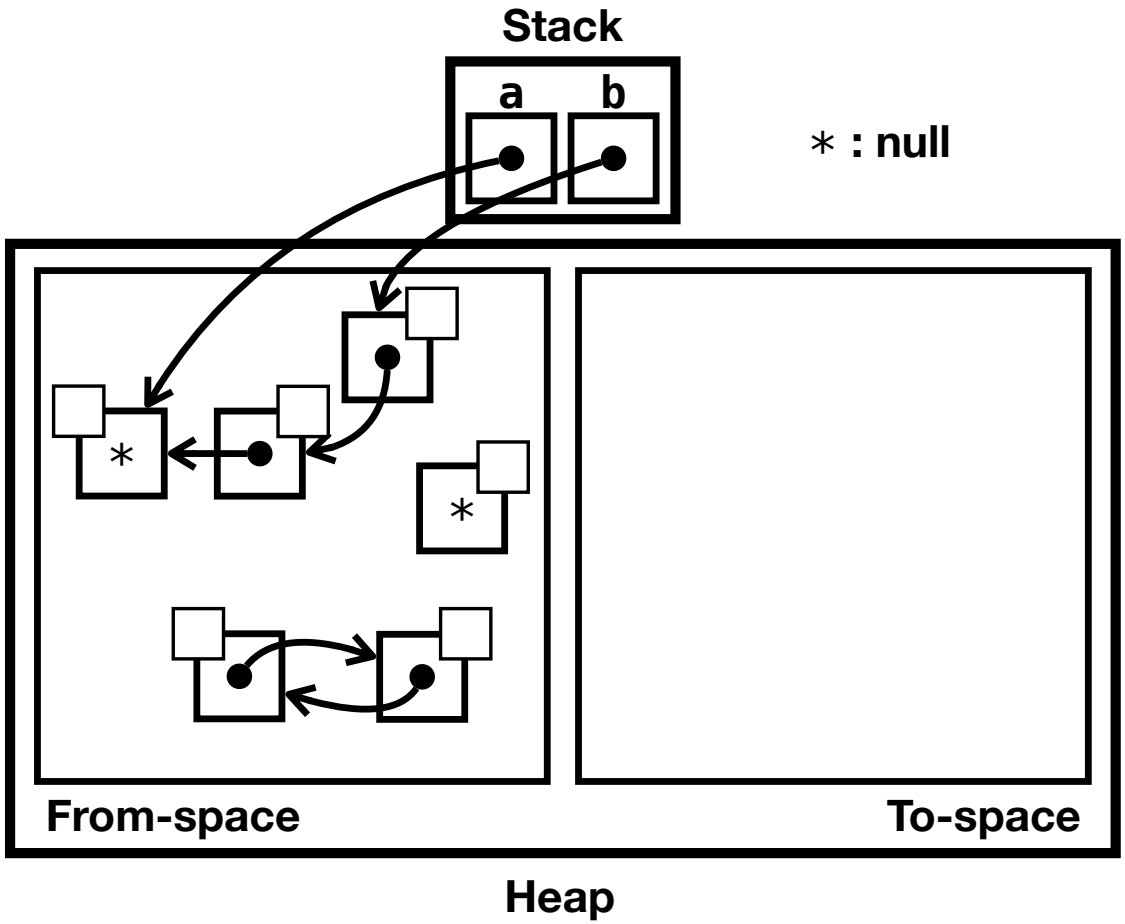
Copying GC (Two-Space GC)

Similar to mark-and-sweep GC, **copying GC** (Two-space GC) is another **tracing GC** algorithm. However, it splits the heap into two spaces, **from-space** and **to-space**, and **copies** all the reachable objects into a **compact** layout:

- **Allocation** – It allocates memory cells only in **from-space**.
- **Deallocation** – It deallocates all the unreachable objects as follows:
  1. **Mark** all memory cells as **unreachable** (white).
  2. **Copy** all memory cells referenced by roots to the **to-space**, leave a **forwarding pointer** in each original cell to its copy, mark the copies as **unscanned** (gray), and **update** the roots to point to the copies.
  3. Repeat until there are no unscanned (gray) memory cells:
     1. **Pick** an unscanned (gray) memory cell in the **to-space**.
     2. **Copy** each memory cell referenced by the picked one to the **to-space** in the same way (gray), unless it already has a forwarding pointer.
     3. **Update** the references of the picked memory cell to point to the copies.
     4. **Mark** the picked memory cell as **scanned** (black).
  4. **Swap** from-space and to-space. All memory cells remaining in the old from-space are **deallocated**.

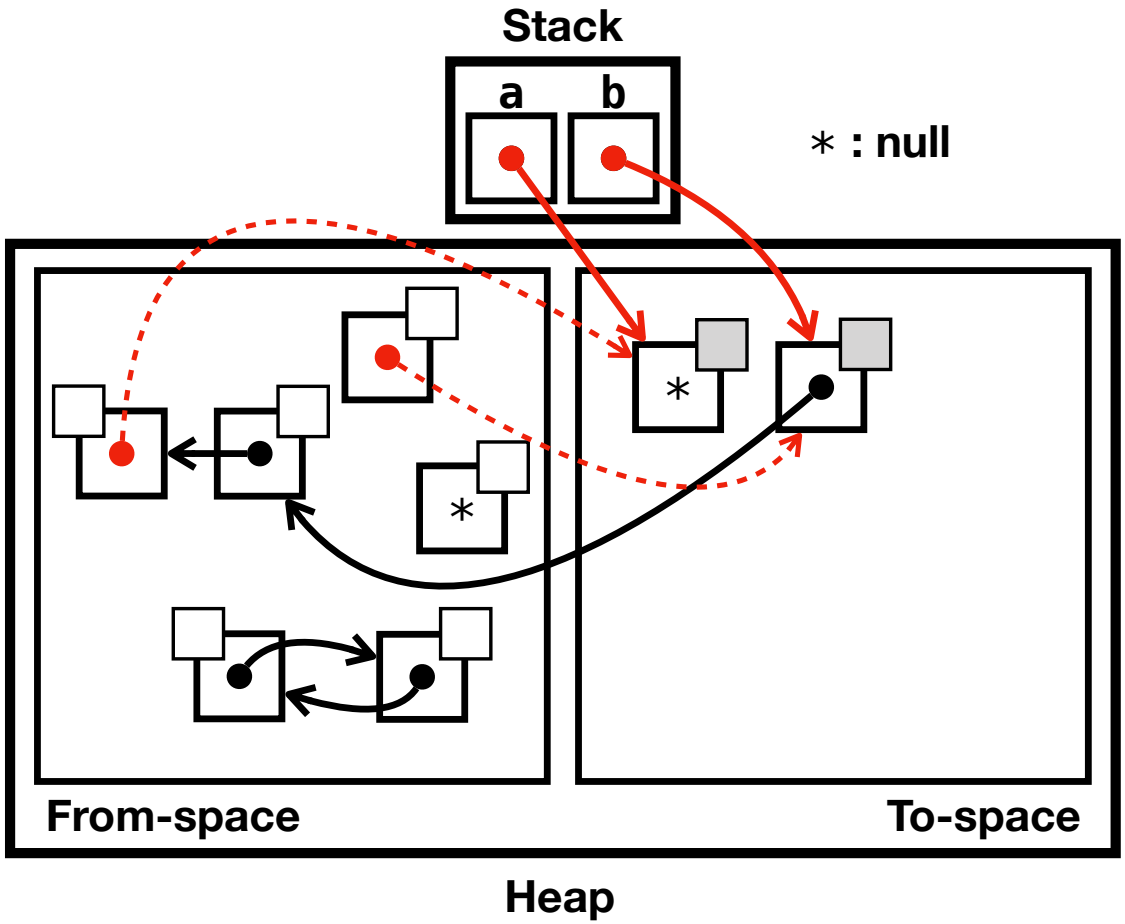
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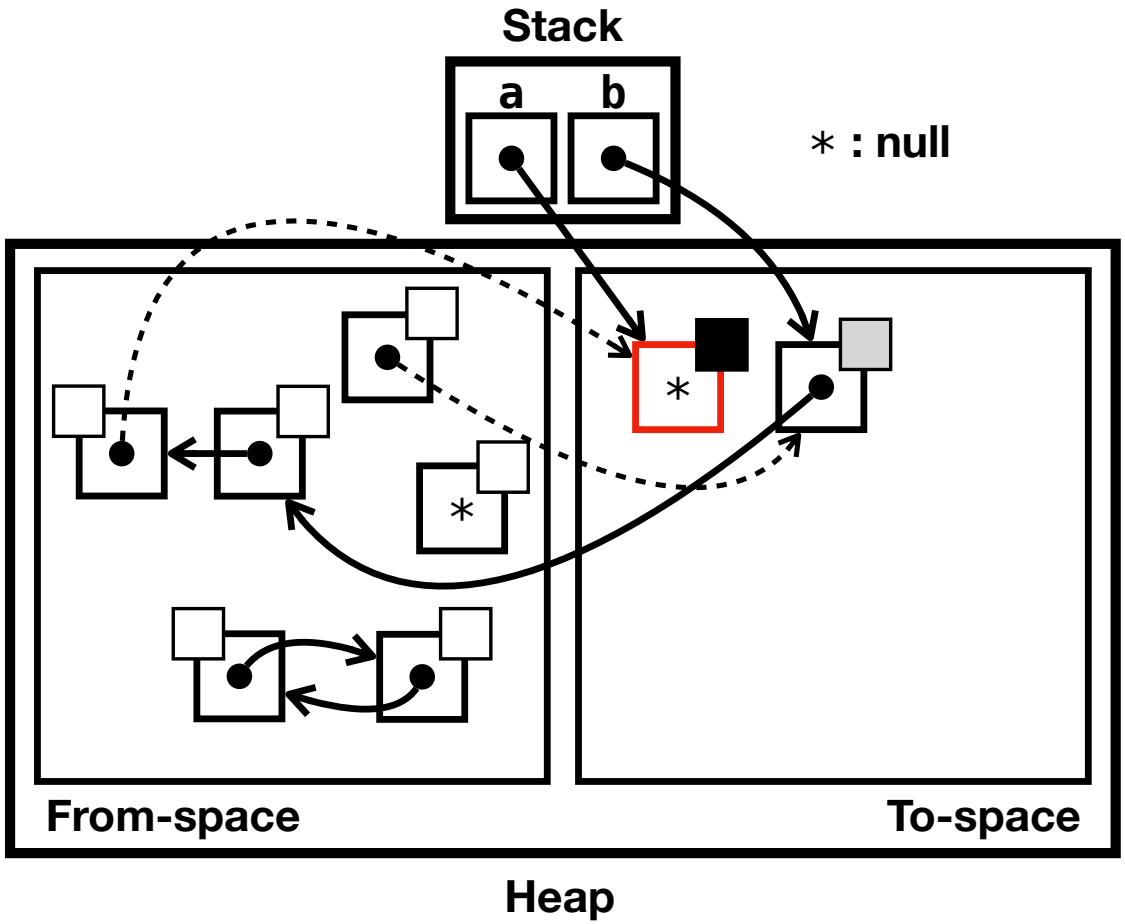
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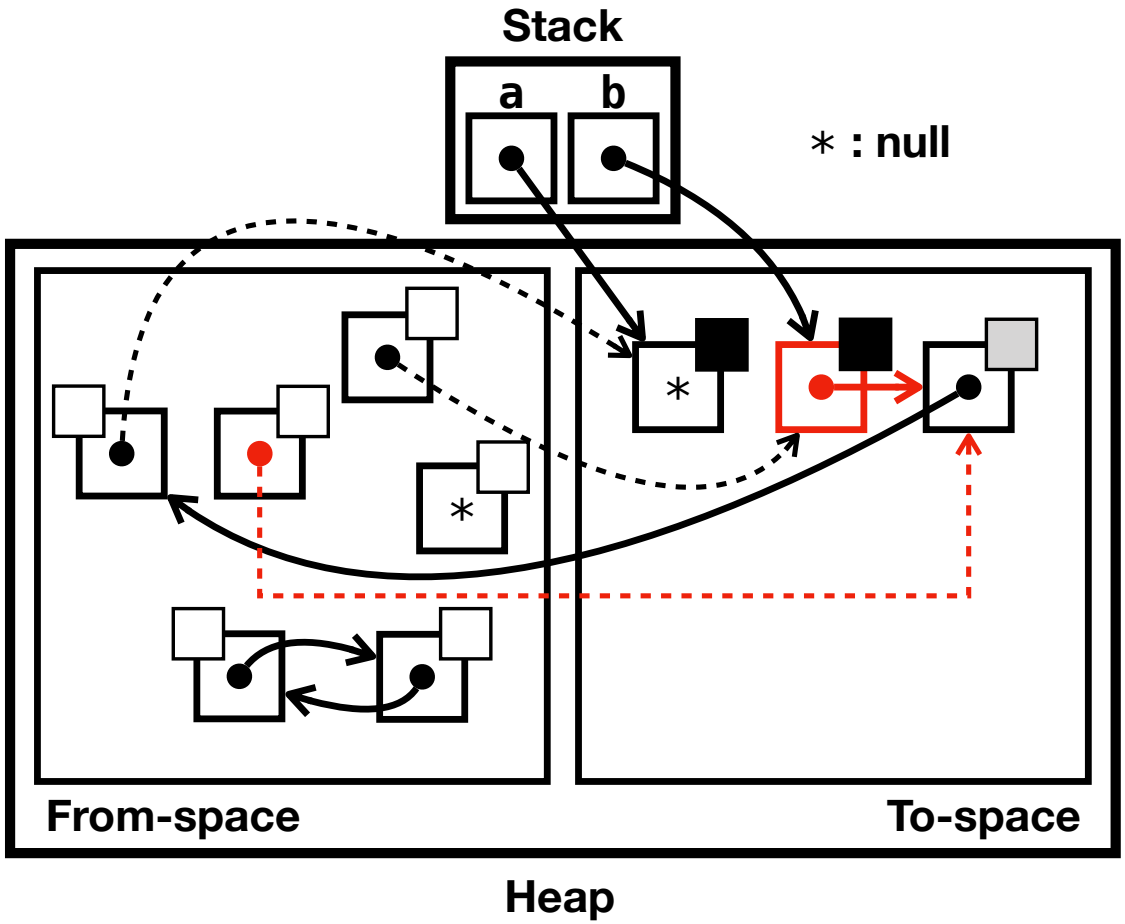
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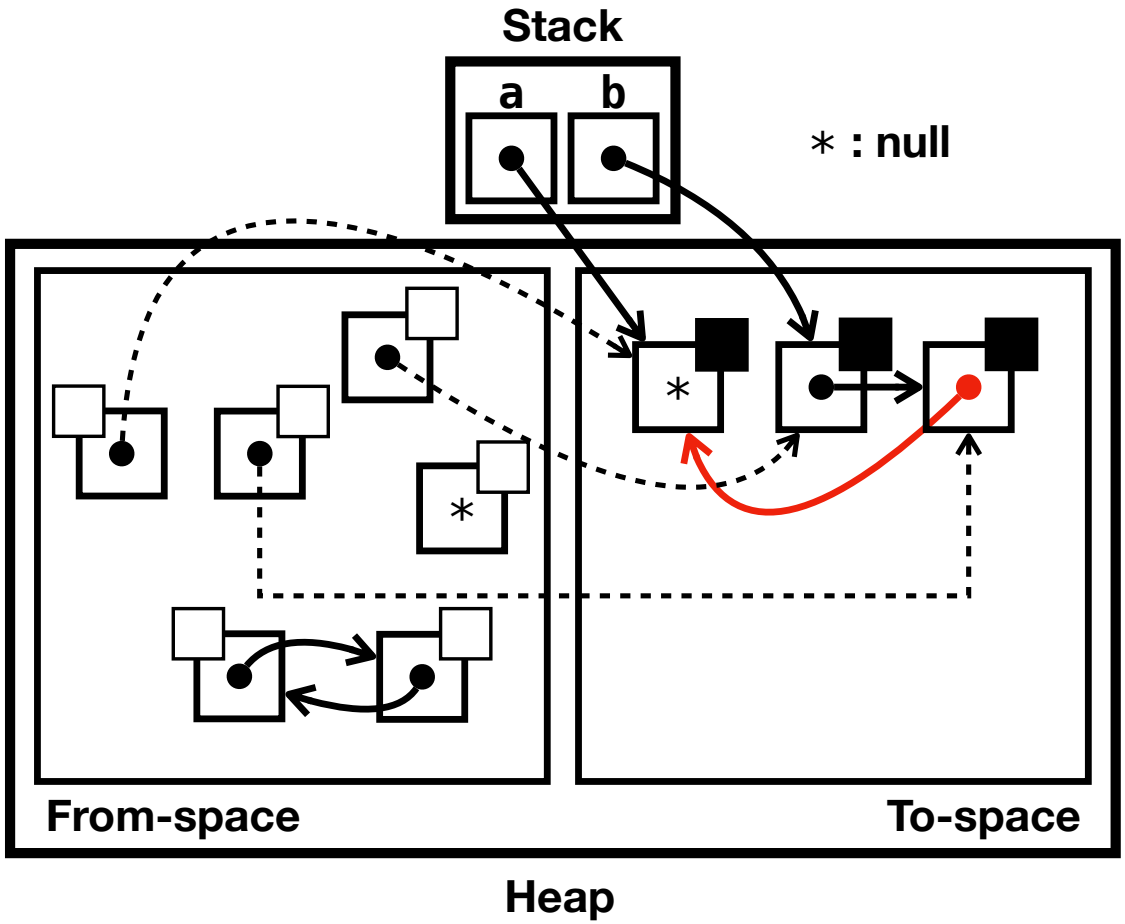
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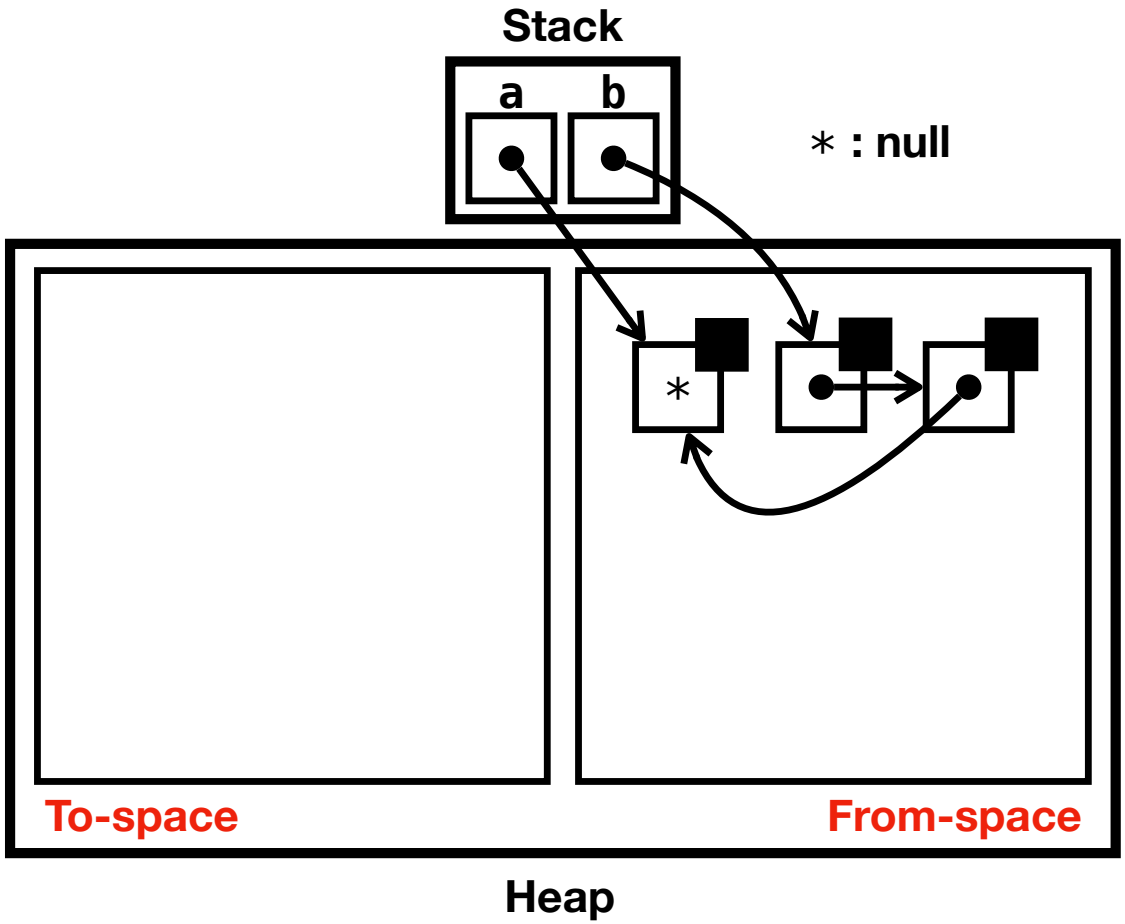
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Pros:

- **Handle Reference Cycles** – It can deallocate unreachable reference cycles.
- No more **Free List** and **Fragmentation** – The heap is always **contiguous** after GC.
- **Fast Allocation** – It only needs to track the first free cell.

Cons:

- **Stop-the-world** – It **stops** the program execution during GC.
- **Only half of the heap** (from-space) is used for allocation.
- **Expensive copying** – It copies all reachable objects on every GC.

Other Techniques

Real-world programming languages use more sophisticated GC algorithms, combine them, or let users choose among them:

- **Generational GC** – Collects **young** objects more often: e.g., Java[1], Python (for reference cycles)[2]
- **Concurrent GC** – Runs GC **concurrently** with the program to reduce pauses: e.g., Java[3], Golang[4]

They also use **compiler optimizations** to reduce heap allocations:

- **Escape Analysis** – Allocates an object on the **stack** if it does not **escape** the function: e.g., Java[5], Golang

Or, a totally different approach, the **ownership** system, is used in Rust.[6]

[1] https://www.oracle.com/webfolder/technetwork/tutorials/obe/java/gc01/index.html
[2] https://github.com/python/cpython/blob/main/InternalDocs/garbage_collector.md
[3] https://docs.oracle.com/en/java/javase/26/gctuning/z-garbage-collector1.html
[4] https://tip.golang.org/doc/gc-guide
[5] https://docs.oracle.com/en/java/javase/26/vm/java-hotspot-virtual-machine-performance-enhancements.html
[6] https://doc.rust-lang.org/book/ch04-01-what-is-ownership.html

Summary

PLRG

- **Stack and Heap** – The stack frame of a function is deleted when the program exits the function, but the heap needs other techniques.
  - **Tail-call optimization (TCO)** discards the current stack frame before a tail call.
- **Manual Memory Management** – Efficient, but error-prone (memory leak, use after free, and double free).
- **Garbage Collection (GC)** – Automatically deallocates unreachable memory cells in heap.
  - **Reference counting** – No pauses, but cannot handle reference cycles.
  - **Mark-and-sweep GC** – Handles reference cycles, but stop-the-world and fragmentation.
  - **Copying GC** – Compact heap and fast allocation, but uses only half of the heap.
  - **Others** – Generational GC, concurrent GC, escape analysis, and ownership.

Next Lecture

- Lazy Evaluation

Jihyeok Park
jihyeok_park@korea.ac.kr
https://plrg.korea.ac.kr